

Project Report Template

Advanced Analytics and Applications

Team 05



Authors:

George Sadik (111111111)

Kushale Fernando (222222222)

Claus Hack (333333333)

Martin Nikoljacic (333333333)

Contact: student@smail.uni-koeln.de

Supervisor: Univ.-Prof. Dr. Wolfgang Ketter

Co-Supervisor: Janik Muires

Chair of Information Systems for Sustainable Society
Faculty of Management, Economics and Social Sciences
University of Cologne

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Table of contents

1	Introduction	1
2	Preprocessing	1
2.1	Support Vector Machines (SVM)	1
2.2	Neural Networks	3
3	Smart Charging Using Reinforcement Learning	3
4	Discussion & Outlook	3
5	Citations and References	3
6	Conclusion	3

List of Figures

1	Absolute error distribution by community area (top 15 by demand) for the recommended configuration (community area, 4h, log1p, RBF).	2
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1 Introduction

This report serves as a template for your team project. It is built using Quarto, which allows you to integrate your Python or R analysis directly into your writing.

By using this template, you ensure that your formatting (margins, line spacing, and fonts) matches the required standards for submission. The left margin is set to 4cm to allow for comments, while the other margins are 2.5cm. Line spacing is set to 1.5, and the main font is Times New Roman.

2 Preprocessing

2.1 Support Vector Machines (SVM)

The SVR model has four dimensions of testing. Two spatial layers (448 census tracts or 77 community areas), target encoding (with or without), the target transformation (normal or log1p) and three temporal resolutions (1h, 4h, 1d). All of those layers are also tested on a linear, polynomial and RBF kernel. We compute all 24 combinations and analyze them in four controlled comparisons, where each phase varies exactly one factor while all others stay fixed. This isolates the effect of every design decision and still yields the overall best SVR as a benchmark against the neural network. The full feature set can be found in Appendix X. Phase 1 compares the two spatial layers at 1h resolution with target encoding and the untransformed target. Phase 2 compares with and without target encoding at 1h across both spatial layers. Target encoding adds one column containing the mean trip count per spatial unit over the training window, so the model learns whether a location is high or low demand¹. Phase 3 compares the untransformed target against log1p on community areas at 1h. Trip counts are strongly right-skewed, so a few high-demand baskets dominate the squared error. Applying log1p compresses these extremes. To avoid selecting a kernel that looks strong on the log scale but fails after back-transformation, the grid search uses a custom RMSE scorer that applies expm1 before computing the error on the original scale. The final phase compares the three temporal resolutions of 1h, 4h and 1d on community areas with target encoding.

Selected: spatial=community_area | target=normal

Table 1: SVR results across four controlled comparisons

Phase str	Configuration str	Spatial str	Bin str	Target str	Target Enc. str	Best str
"1: spatial"	"census_tract_1h"	"census_tract"	"1h"	"normal"	"yes"	"Pol"
"1: spatial"	"community_area_1h"	"community_area"	"1h"	"normal"	"yes"	"Pol"
"2: target"	"community_area_1h"	"community_area"	"1h"	"normal"	"yes"	"Pol"
"2: target"	"community_area_1h + log1p"	"community_area"	"1h"	"log1p"	"yes"	"RB"
"3: temporal"	"community_area_1h"	"community_area"	"1h"	"normal"	"yes"	"Pol"
"3: temporal"	"community_area_4h"	"community_area"	"4h"	"normal"	"yes"	"Pol"
"3: temporal"	"community_area_1d"	"community_area"	"1d"	"normal"	"yes"	"Pol"

¹Dummy encoding was rejected because it would add up to 448 sparse columns with high computational cost for SVR's quadratic kernel scaling.

Community areas consistently outperform census tracts (R^2 0.84 vs 0.65). Target encoding is critical for census tracts (R^2 0.65 vs 0.24) but marginal for community areas (0.84 vs 0.83). On the 1h scale, normal and log1p perform nearly identically (R^2 0.84 vs 0.83), but at 4h log1p clearly wins (0.91 vs 0.88). Wider time windows improve R^2 (0.84 $\rightarrow$ 0.88 $\rightarrow$ 0.93) but lose the intra-day detail a fleet operator needs. Across all phases, the polynomial kernel achieves the best results on the untransformed target, while the RBF kernel wins whenever log1p is applied. The best model by R^2 is community area at daily resolution with target encoding and a polynomial kernel (R^2 0.93, nMAE 0.24). However, a daily total per neighborhood is too coarse for a fleet operator to actually work with. While wider time windows perform better, they are not operationally suitable as the operator cannot coordinate the fleet precisely enough. Considering both predictive quality and operational usefulness, community area at 4h resolution with target encoding, log1p and an RBF kernel is the better choice (R^2 0.91, nMAE 0.30)². We considered recommending census tracts for finer spatial detail but decided against it for two reasons. First, fewer than half of the records contain census-tract information, so the model is trained on a significantly reduced dataset at that resolution (see chapter X). Second, community areas are already small enough to be operationally useful. The average community area covers roughly three square miles, meaning a taxi can cross one in under fifteen minutes at a typical city speed of 20 mph. Community-level forecasts translate directly into vehicle allocations, whereas many individual census tracts generate fewer than one trip per window, a granularity too fine to plan vehicles against.

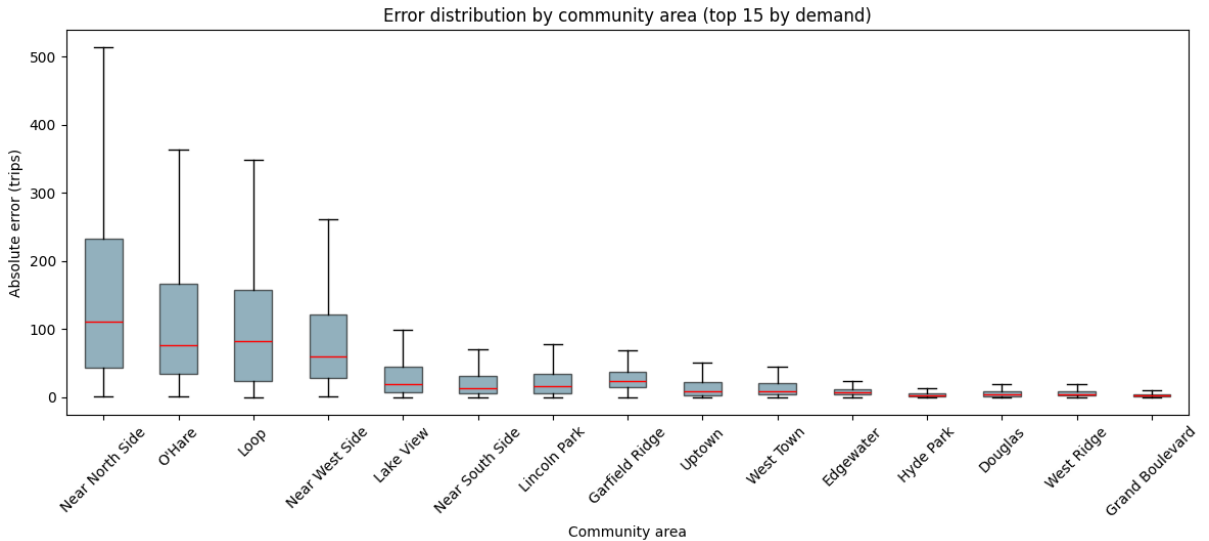


Figure 1: Absolute error distribution by community area (top 15 by demand) for the recommended configuration (community area, 4h, log1p, RBF).

²Due to different scales across temporal resolutions, the normalized MAE (MAE / mean demand) is used to enable a valid comparison.

2.2 Neural Networks

3 Smart Charging Using Reinforcement Learning

4 Discussion & Outlook

5 Citations and References

You can easily cite your sources using BibLaTeX. For example, you can cite the Quarto documentation (Team, 2024) or a textbook like “R for Data Science” (Wickham et al., 2023). The bibliography will be automatically generated at the end of the document.

6 Conclusion

The conclusion should summarize your findings and suggest future work.

References

- Team, Q. (2024). Quarto: An open-source scientific and technical publishing system. *Journal of Modern Data Science*. <https://quarto.org>
- Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *R for data science*. O'Reilly Media.